

Solution Requirements

Date	27 June 2026
Team ID	SWTID-2026-9097
Project Name	Voice--Based--Concept--Understanding--Analyzer
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register and log in using email/password or institution credentials before accessing the system.	High
2	Authorization Levels	Student: Upload/record audio, view reports. Admin: Manage users, topics, evaluation data, and monitor system usage.	High
3	External Interfaces	Integrates with Whisper API for speech-to-text, Gemini API for AI feedback, Sentence-BERT for semantic similarity, and Librosa for audio feature extraction.	High
4	Transaction Processing	Accepts audio recordings, processes speech into text, evaluates conceptual understanding, calculates scores, stores results, and generates reports.	High
5	Reporting	Generates evaluation reports containing transcript, semantic similarity score, fluency score, waveform visualization, AI feedback, and downloadable PDF reports.	High
6	Business Rules	Audio must be in supported formats (WAV/MP3). Maximum recording duration is limited. Scores are calculated using semantic similarity and speech quality metrics. Only authenticated users can access reports.	High

S.No	Requirement Category	Requirement Description	Priority
7	Compliance to Laws or Regulations	User data is stored securely following general data privacy guidelines. Audio and evaluation reports are accessible only to authorized users.	Medium
8	Other	Users can view previous evaluations, compare performance across attempts, and download historical reports.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Audio transcription and evaluation should complete quickly after upload.	Complete analysis within 15–30 seconds for a 2-minute audio.
2	Scalability	The system should support multiple users without significant performance degradation.	Supports 100+ concurrent users.
3	Security & Data Privacy	User credentials and evaluation reports must be protected using secure authentication and encrypted communication.	Password hashing, HTTPS communication, secure data storage.
4	Reliability & Availability	The application should be consistently available and recover gracefully from failures.	99% system availability with proper error handling.
5	Usability & Accessibility	Interface should be intuitive, responsive, and easy to use on desktop and mobile devices.	Responsive UI with simple navigation and accessibility-friendly design.
6	Maintainability & Portability	Code should be modular, well-documented, and deployable on Windows, Linux, or cloud platforms.	Modular architecture with reusable components and deployment support on multiple platforms.

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